

Quantum Optimization for Node Placement in Device-Free Localization

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Abstract—Device-Free Localization (DFL) enables the detection and tracking of targets without requiring them to carry electronic devices. Its performance strongly depends on the placement of signal sources (Access Points, APs) and receivers (Monitoring Points, MPs), as each AP–MP pair (link) contributes differently to localization accuracy. Hence, given a large number of candidate locations for APs and MPs, the number of possible links grows combinatorially, making it challenging to select an optimal subset under APs and MPs budget constraints. As the number of candidate positions increases, exhaustive search quickly becomes impractical.

In this paper, we propose a quantum optimization approach for node placement in Device-Free Localization. We formulate the problem as a Quadratic Unconstrained Binary Optimization (QUBO) model defined over all possible AP–MP links, where binary variables indicate whether a link is selected. The objective function balances two goals: maximizing the contribution of individual links to localization accuracy and reducing redundancy among selected links, thereby promoting both coverage and information diversity. To respect practical hardware constraints on the number of APs and MPs, we propose an efficient binary search strategy that iteratively guides the QUBO toward feasible solutions under given budgets. We solve the resulting QUBO instances using quantum annealing on a D-Wave system.

We evaluate our approach in a realistic indoor environment and demonstrate that quantum annealing achieves localization accuracy comparable to state-of-the-art classical methods, while providing up to a $129\times$ speedup. These results highlight the potential of quantum computing as a viable tool for efficient and scalable DFL system design.

Index Terms—QUBO, quantum annealing, sensor node placement, device-free localization, binary search, time-to-solution

I. INTRODUCTION

Device-Free localization (DFL) enables tracking and positioning of targets without requiring them to carry any electronic device, making it particularly attractive for applications such as object detection [?], intrusion monitoring [?], and smart-building occupancy estimation [?], among others [1]. A typical DFL system operates in two phases: an offline training phase and an online inference phase. In the offline phase, for each of k candidate target locations, the received signal strength (RSS) from n transmitters (Access Points, APs) is recorded at m receivers (Monitoring Points, MPs), forming a fingerprint database over all $n \times m$ AP–MP links. In the online phase, the system measures an RSS vector corresponding to an unknown target location and estimates its position by matching

it to the closest fingerprint in the database. Each AP–MP pair defines a communication link whose RSS is perturbed when a target intersects its Fresnel zone [2], enabling localization without any active participation from the target.

Despite its simplicity, the performance of a DFL system is fundamentally governed by the placement of the n Access Points (APs) and m Monitoring Points (MPs) selected from a set of N candidate locations. These placements implicitly define a set of potential AP–MP links, where each possible pairing between candidate AP and MP locations yields up to N^2 potential links. From this large pool of candidate links, the system must select an optimal subset that maximizes localization accuracy. Importantly, not all links contribute equally: effective links should span the monitored area with diverse orientations and cover complementary spatial regions, producing RSS measurements that respond differently to target movement. Such diversity provides rich, non-overlapping information that improves DFL precision. On the other hand, spatially correlated or parallel links often exhibit similar responses, leading to redundancy that adds redundant information while consuming the limited hardware budget of n APs and m MPs. Consequently, DFL node placement is considered as a combinatorial link selection problem driven by link quality and diversity.

Selecting an optimal subset of links under hardware constraints on the number of APs and MPs is computationally challenging. Given N candidate locations for APs and MPs, the space of possible AP–MP links scales as N^2 , and selecting a subset while simultaneously enforcing separate budgets on distinct AP and MP nodes leads to an exponentially large search space. Exhaustive search is therefore infeasible in practical settings. Moreover, standard optimization approaches struggle to enforce these coupled constraints effectively [3], [4].

In this paper, we propose a quantum optimization framework for node placement in DFL by formulating the problem as a QUBO defined over *all possible AP–MP links*. Each binary variable represents whether a specific link, formed by pairing a candidate AP location with a candidate MP location, is selected. Our objective function captures two key properties derived from RSS fingerprints: (i) *link importance*, which quantifies how informative a link is for distinguishing between

locations, and (ii) *link redundancy*, which measures the degree of similarity between links. By maximizing importance while penalizing redundancy, the formulation naturally promotes diverse and informative link sets.

To enforce practical constraints on the number of deployed APs and MPs, we introduce a penalty-free mechanism based on a single scalar parameter α that controls the trade-off between importance and redundancy. We prove that the number of selected APs and MPs is monotonically non-decreasing in α , enabling an efficient binary search procedure that identifies the optimal feasible deployment without requiring manual parameter tuning. The resulting QUBO instances are solved using quantum annealing on D-Wave hardware, leveraging its native capability for large-scale binary optimization.

The contributions of this paper are summarized as follows:

- A QUBO formulation for AP and MP node placement in DFL that jointly captures link importance and inter-link redundancy from RSS fingerprint data, where each binary variable represents a specific AP candidate location paired with a specific MP candidate location.
- A penalty-free budget enforcement mechanism based on a single scalar α , with a formal monotonicity proof guaranteeing that binary search over α finds the optimal feasible deployment (Section III-E).
- An adaptive binary search algorithm (Algorithm 1) that enforces simultaneous AP and MP budgets and retains the best feasible deployment across all iterations.
- Results of deploying the proposed technique in a real indoor environment, demonstrating that quantum annealing achieves comparable localization accuracy to classical simulated annealing while delivering $129\times$ speedup.

The rest of this paper is structured as follows. Section II reviews background on DFL and quantum annealing. Section III presents the details of the proposed technique. Section IV describes the experimental setup and results. Section V surveys related work. Finally, Section VI concludes the paper.

II. BACKGROUND

In this section, we give a quick background about device-free localization and quantum annealing.

A. Device-Free Passive Localization

Device-free localization (DFL) is a sensing paradigm in which the position or movement of a person is inferred from changes in ambient RF signals, with no device carried by the subject [1]. Because the monitored individual plays no active role, DFL is particularly well-suited for scenarios where carrying a device is not feasible, such as elderly care, intrusion detection, and emergency response.

Several RF-based technologies have been explored for DFL, including WiFi Channel State Information (CSI) [5] and Radio Tomographic Imaging (RTI) [2]. In RTI and other link-based DFL systems, a set of N sensor nodes is deployed around the monitored area. Each node acts as either a transmitter (AP)

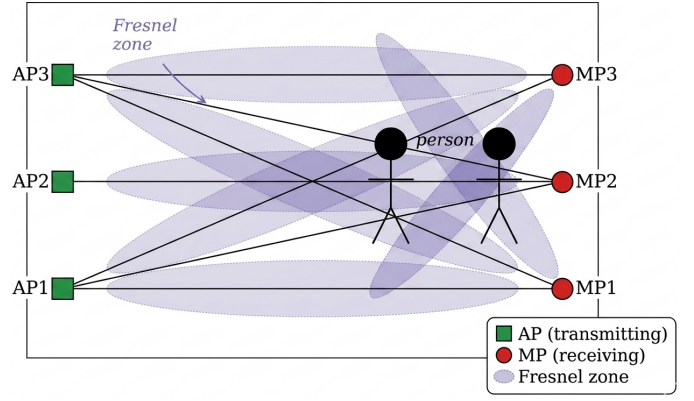


Fig. 1. Illustrative RTI deployment. AP nodes (transmitters) and MP nodes (receivers) form directed links. Each link's Fresnel zone is the elliptical region between its endpoints; a person crossing this zone causes a measurable RSS drop that indicates their position.

or a receiver (MP), forming up to $N(N - 1)$ directed AP-MP links. When a person moves through the area, their body blocks or deflects RF energy along links whose *Fresnel zones* it intersects. The Fresnel zone of a link is an ellipsoidal region centered on the line-of-sight path between AP and MP: any obstacle penetrating this zone causes a measurable drop in the received RSS. Fig. 1 illustrates a typical deployment with AP and MP nodes, links, and the elliptical Fresnel zones that each link contributes to sensing.

The quality of the localization system depends critically on which links are active, not merely how many. A useful measure of individual link quality is the temporal RSS variance, $\sigma_{ij}^2 = \text{Var}_t[\text{RSS}_{ij}(t)]$. Links with high variance are frequently disrupted by a moving person and carry strong localization information, while links with low variance traverse static regions and contribute little. Beyond individual link quality, the collective diversity of the active link set matters: two links that respond identically to all movements are redundant, and selecting both wastes budget that could fund a more informative link. Good node placement therefore maximizes both individual link quality and collective spatial diversity across the deployed set.

B. Quantum Annealing

Quantum annealing is a computational paradigm designed for combinatorial optimization, seeking the lowest-energy configuration among an exponentially large set of candidates [6], [7]. It is rooted in adiabatic quantum computation [8], [9]: a quantum system is initialized in the ground state of a simple Hamiltonian H_0 and slowly evolved toward a problem Hamiltonian H_f that encodes the optimization objective [10], [11]. When the evolution is sufficiently slow, the adiabatic theorem guarantees that the system stays in its ground state and arrives at the optimal solution.

The key advantage over classical search is quantum tunneling and superposition [12]: instead of climbing over energy barriers as in simulated annealing, the system can tunnel

through them, which helps it escape local minima. The time evolution follows the Schrödinger equation [13]:

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = H(t) |\psi(t)\rangle \quad (1)$$

where $H(t)$ interpolates between H_0 and H_f through a schedule parameter $s(t) \in [0, 1]$ [14]:

$$H(t) = (1 - s(t))H_0 + s(t)H_f \quad (2)$$

Quantum annealers are well matched to Quadratic Unconstrained Binary Optimization (QUBO) problems [15], [16]:

$$\text{minimize } f(\mathbf{x}) = \sum_{i,j} Q_{ij} x_i x_j + \sum_i P_i x_i \quad (3)$$

where $\mathbf{x} \in \{0, 1\}^n$, Q_{ij} are quadratic coefficients, and P_i are linear coefficients [17]. A QUBO instance maps directly onto an Ising Hamiltonian, where binary variables correspond to spin states and the objective becomes the energy function [18]–[21]. This equivalence lets quantum annealers such as D-Wave systems solve QUBO instances natively [22], [23]. Quantum annealing has been applied across finance [24], logistics [25], machine learning [26], and network optimization [27].

III. METHODOLOGY

A. Problem Formulation

Let $\mathcal{N} = \{0, \dots, N-1\}$ denote a pool of N candidate sensor node positions in the monitored area. Each node can operate either as an Access Point (AP, transmitter) or a Monitor Point (MP, receiver). A deployment consists of selecting directed AP-MP pairs (i, j) with $i \neq j$; there are $P = N(N-1)$ such pairs in total.

All pairs are enumerated in a fixed order and the deployment is represented as a binary vector $\mathbf{x} \in \{0, 1\}^P$, where $x_p = 1$ means the directed pair indexed by p is active. Let K denote the number of reference fingerprint locations in the training database, and let $\text{RSS}(k, i, j)$ denote the channel-averaged Received Signal Strength (dBm) at fingerprint k when AP i transmits to MP j .

The deployment is subject to two hardware budgets:

$$\text{AP_count}(\mathbf{x}) = |\{i : \exists j, x_{(i,j)} = 1\}| \leq n, \quad (4)$$

$$\text{MP_count}(\mathbf{x}) = |\{j : \exists i, x_{(i,j)} = 1\}| \leq m, \quad (5)$$

where n and m are the maximum numbers of distinct AP and MP nodes allowed, respectively. Table I summarises all notation used throughout the paper.

B. Link-Level Metrics

Before presenting the QUBO formulation, we define two link-level properties that jointly characterize the value of an AP-MP link for localization.

A link is **important** if its RSS changes substantially depending on where the person is standing. Concretely, a link whose RSS stays almost constant across all fingerprint locations means the link is insensitive to movement and carries very little useful information, regardless of how strong the signal is. A link whose RSS varies widely across locations means

TABLE I
SUMMARY OF NOTATION

Symbol	Description
N	Number of candidate sensor node locations.
$P = N(N-1)$	Total number of directed AP-MP pairs.
K	Number of fingerprint locations in the training database.
i, j	Node indices, $i \neq j$.
p, q	Pair indices in $\{0, \dots, P-1\}$, each mapping to a unique directed pair (i, j) .
$\mathbf{x} \in \{0, 1\}^P$	Binary decision vector; $x_p = 1$ means pair p is active.
$\text{RSS}(k, i, j)$	Received Signal Strength (dBm) at fingerprint k , AP i , MP j .
I_p	Importance of pair p : RSS variance across K locations.
R_{pq}	Redundancy between pairs p and q : absolute Pearson correlation of their RSS vectors, $R_{pq} \in [0, 1]$.
$\alpha \in [0, 1]$	Trade-off scalar; higher α favours importance.
n	Maximum number of AP nodes allowed (AP budget).
m	Maximum number of MP nodes allowed (MP budget).
ε	Termination threshold for the binary search interval.
a, b	Current lower and upper bounds of the binary search.
err	KNN localization error (m) of a feasible deployment.
best_err	Lowest localization error seen across feasible iterations.
α^*	Largest feasible α found by binary search.
$\mathbf{x}^*$	Optimal deployment: feasible solution with lowest err.

that different positions produce different measurements on that link, which is exactly what a localizer needs. We formalize this as the variance of the link's RSS vector across the K fingerprint locations:

$$I_p = \text{Var}_k [\text{RSS}(k, i, j)]. \quad (6)$$

High I_p indicates that pair $p = (i, j)$ is sensitive to position and worth including in the deployment.

Two links are **redundant** if they respond similarly to the person's movement across all fingerprint locations. If two links always increase or decrease their RSS together, selecting both gives no more information than selecting either one alone, while consuming twice the budget. We measure this with the absolute Pearson correlation between the two links' RSS vectors:

$$R_{pq} = |\text{corr}([\text{RSS}(k, i, j)]_k, [\text{RSS}(k, i', j')]_k)|. \quad (7)$$

$R_{pq} = 1$ means the two links carry identical positional information; $R_{pq} = 0$ means they are completely independent. Both I_p and R_{pq} are computed directly from the RSS fingerprint database and require no additional sensing beyond what the localization system already collects.

C. QUBO Formulation

With importance and redundancy defined, the link selection problem can be written as a single objective that rewards informative links and penalizes redundant ones. We want to select a set of pairs that has high total importance (so the deployment is sensitive to movement) while keeping pairwise redundancy low (so the selected links collectively cover the space well). These two goals are in tension: maximizing importance alone tends to select many links, including correlated ones, while minimizing redundancy alone pushes toward a sparse, diverse set that may miss individually strong links. The scalar $\alpha \in [0, 1]$ governs this trade-off:

$$Q(\mathbf{x}, \alpha) = -\alpha \sum_p I_p x_p + (1 - \alpha) \sum_{p \neq q} R_{pq} x_p x_q. \quad (8)$$

The first term is negative, so minimizing Q rewards selecting pairs with high importance. The second term is positive, so minimizing Q penalizes selecting two pairs that are highly correlated. When $\alpha = 1$ the formulation reduces to pure importance maximization; when $\alpha = 0$ it reduces to pure redundancy minimization. At intermediate values, the solver balances both and typically finds a compact, diverse deployment that outperforms either extreme. The optimal deployment for a given α is:

$$\mathbf{x}^* = \arg \min_{\mathbf{x} \in \{0,1\}^P} Q(\mathbf{x}, \alpha). \quad (9)$$

Importantly, no penalty term for the node budget appears in (8). The budget is enforced entirely by the binary search over α described next, without any coefficient to tune.

D. Adaptive Binary Search for Budget Enforcement

The key observation is that as α increases, the importance term gains more weight and the unconstrained solution tends to select more pairs, which in turn activates more distinct AP and MP nodes. We formalize this in Section III-E: AP_count and MP_count of the optimal solution are both non-decreasing functions of α .

This monotonicity property is what makes budget enforcement possible without a penalty term. Because the node counts are monotone in α , the set of feasible values of α (those whose optimal solution satisfies both budgets) forms a contiguous interval $[0, \alpha_{\max}]$. We want the largest feasible α , because a higher α means the importance term has more influence, which generally yields a better deployment. Finding $\alpha_{\max}$ is therefore a standard binary search problem.

Algorithm 1 implements this search. It maintains an interval $[a, b]$ where a is always feasible and b is always infeasible, and at each iteration it evaluates the midpoint. If the midpoint is feasible, it raises the lower bound and records the localization error of the current deployment. If the midpoint is infeasible (too many distinct nodes are activated), it lowers the upper bound. After $\lceil \log_2(1/\varepsilon) \rceil$ iterations the interval shrinks below ε and the algorithm returns the best feasible deployment seen across all iterations, not just the final one. With $\varepsilon = 0.01$, this requires exactly 7 QUBO solves.

Algorithm 1 Binary Search for Optimal α^*

Input: Importance I , redundancy R , AP budget n , MP budget m , threshold ε

Output: Optimal scalar α^* , selected pairs $\mathbf{x}^*$

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1:  $a \leftarrow 0.0$ ,  $b \leftarrow 1.0$ ,  $\mathbf{x}^* \leftarrow \text{None}$ ,  $\text{best\_err} \leftarrow \infty$ 
2: while  $b - a \geq \varepsilon$  do
3:    $\alpha \leftarrow (a + b)/2$ 
4:    $\tilde{\mathbf{x}} \leftarrow \arg \min_{\mathbf{x}} Q(\mathbf{x}, \alpha)$  ▷ Solve QUBO with
   quantum or classical annealer
5:   Compute AP_count( $\tilde{\mathbf{x}}$ ) and MP_count( $\tilde{\mathbf{x}}$ )
6:   if AP_count  $\leq n$  and MP_count  $\leq m$  then
7:      $a \leftarrow \alpha$  ▷ Feasible: push lower bound up
8:     err  $\leftarrow$  Localizer( $\tilde{\mathbf{x}}$ ) ▷ KNN error, evaluated only
   when feasible
9:     if err  $<$  best_err then
10:       $\mathbf{x}^* \leftarrow \tilde{\mathbf{x}}$ , best_err  $\leftarrow$  err
11:     end if
12:   else
13:      $b \leftarrow \alpha$  ▷ Over budget: push upper bound down
14:   end if
15: end while
16:  $\alpha^* \leftarrow a$  return  $\alpha^*$ ,  $\mathbf{x}^*$ 

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E. Monotonicity of Optimal Node Counts

We now prove that the binary search in Algorithm 1 is provably correct. The key result is that as α increases, the number of distinct AP and MP nodes activated by the unconstrained optimum of (8) is non-decreasing. This guarantees that the feasible region in α -space is an interval, which is the necessary and sufficient condition for binary search to work.

Setup. Fix all importance values $\{I_p\}$ and redundancy values $\{R_{pq}\}$. For any $\alpha \in [0, 1]$, let $\mathbf{x}^*(\alpha) \in \arg \min_{\mathbf{x}} Q(\mathbf{x}, \alpha)$ denote any optimal solution (breaking ties lexicographically). Define:

$$A(\alpha) = |\{i : \exists j, x_{(i,j)}^*(\alpha) = 1\}| \quad (\text{AP count}), \quad (10)$$

$$M(\alpha) = |\{j : \exists i, x_{(i,j)}^*(\alpha) = 1\}| \quad (\text{MP count}). \quad (11)$$

Lemma 1. For any directed pair p , if $\alpha_1 < \alpha_2$ and $x_p^*(\alpha_1) = 1$, then there exists an optimal solution at α_2 with $x_p^*(\alpha_2) = 1$.

Proof. Let $\mathbf{x}^1 = \mathbf{x}^*(\alpha_1)$ with $x_p^1 = 1$, and let $\mathbf{x}^2 = \mathbf{x}^*(\alpha_2)$. If $x_p^2 = 1$ we are done. Suppose for contradiction that $x_p^2 = 0$. By optimality:

$$Q(\mathbf{x}^1, \alpha_1) \leq Q(\mathbf{x}^2, \alpha_1), \quad (12)$$

$$Q(\mathbf{x}^2, \alpha_2) \leq Q(\mathbf{x}^1, \alpha_2). \quad (13)$$

Write $Q(\mathbf{x}, \alpha) = \alpha f(\mathbf{x}) + g(\mathbf{x})$ where $f(\mathbf{x}) = -\sum_p I_p x_p$ and $g(\mathbf{x}) = \sum_{p \neq q} R_{pq} x_p x_q$. Adding (12) and (13) and substituting:

$$(\alpha_2 - \alpha_1)[f(\mathbf{x}^2) - f(\mathbf{x}^1)] \leq 0. \quad (14)$$

Since $\alpha_2 > \alpha_1$, this gives $f(\mathbf{x}^2) \leq f(\mathbf{x}^1)$, i.e., $\sum_p I_p x_p^1 \leq \sum_p I_p x_p^2$. Now construct $\hat{\mathbf{x}}$ from $\mathbf{x}^2$ by setting $\hat{x}_p = 1$ and

keeping all other coordinates unchanged. The marginal gain of activating pair p in the objective is:

$$Q(\hat{x}, \alpha) - Q(x^2, \alpha) = -\alpha I_p + 2(1 - \alpha) \sum_{q \neq p} R_{pq} x_q^2. \quad (15)$$

This quantity is non-positive at α_1 (since x^1 with $x_p^1 = 1$ is optimal at α_1). Because the term $-\alpha I_p$ becomes more negative as α increases (with $I_p \geq 0$), the marginal gain of activating p is at least as large at α_2 as at α_1 . Therefore $Q(\hat{x}, \alpha_2) \leq Q(x^2, \alpha_2)$, contradicting the optimality of x^2 with $x_p^2 = 0$. $\square$

Theorem 1. $A(\alpha)$ and $M(\alpha)$ are non-decreasing on $[0, 1]$.

Proof. Let $\alpha_1 < \alpha_2$. By Lemma 1, every pair active at α_1 is also active in some optimal solution at α_2 . The active pair set at α_2 is therefore a superset of that at α_1 . Every AP (resp. MP) node that appears at α_1 also appears at α_2 , so $A(\alpha_1) \leq A(\alpha_2)$ and $M(\alpha_1) \leq M(\alpha_2)$. $\square$

Correctness of binary search. Theorem 1 implies that the feasible set $\mathcal{F} = \{\alpha : A(\alpha) \leq n, M(\alpha) \leq m\}$ is an interval $[0, \alpha_{\max}]$. Algorithm 1 maintains $a \in \mathcal{F}$ and $b \notin \mathcal{F}$ throughout, halving $[a, b]$ at each step. After $\lceil \log_2(1/\varepsilon) \rceil$ iterations the interval width is below ε , and a is a provably correct approximation to $\alpha_{\max}$.

IV. EVALUATION

A. Experimental Setup

Experiments are conducted on the RTI Indoor Basement dataset [28], which provides RSS measurements collected from $N = 10$ sensor nodes across $K = 21$ reference fingerprint locations, yielding $P = N(N - 1) = 90$ directed AP-MP pairs. AP and MP budgets are set to $n = m = 5$. The binary search termination threshold is $\varepsilon = 0.01$, which bounds the number of iterations to $\lceil \log_2(1/\varepsilon) \rceil = 7$.

Two solvers are evaluated and compared. **Simulated Annealing (SA):** using 1000 sweeps and 100 reads per QUBO solve; the per-solve time τ is the measured wall-clock elapsed time. **Quantum Annealing (QA):** with a hardware anneal time of $\tau = 20 \mu\text{s}$ per solve; all QA numbers are theoretical projections and were not obtained on physical quantum hardware.

Localization is evaluated using a k -Nearest Neighbour (KNN) classifier with $k = 3$, computing Euclidean distance over the RSS vectors of the selected pairs. The Time-to-Solution at 99% success probability is:

$$\text{TTS}_{99} = \tau \cdot \frac{\ln(1 - 0.99)}{\ln(1 - p_s)}, \quad (16)$$

where p_s is the empirical success probability over all reads. When $p_s = 1.0$, a single run suffices and $\text{TTS}_{99} = \tau$. Each solver is evaluated over 100 independent runs using seeds 0–99 to obtain statistically meaningful results.

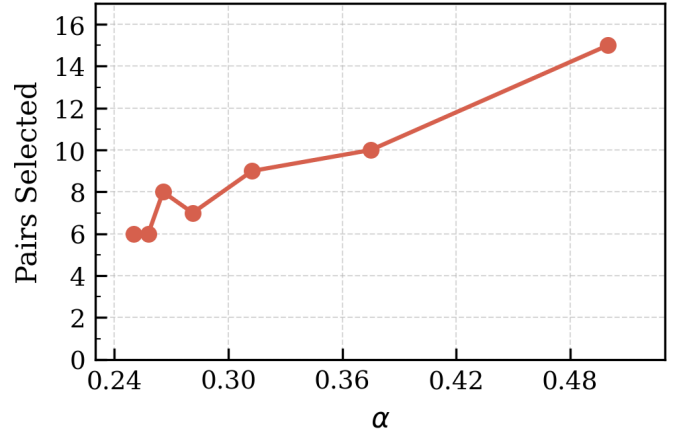


Fig. 2. Total active pairs as a function of α (QA solver). The count rises monotonically, consistent with Theorem 1.

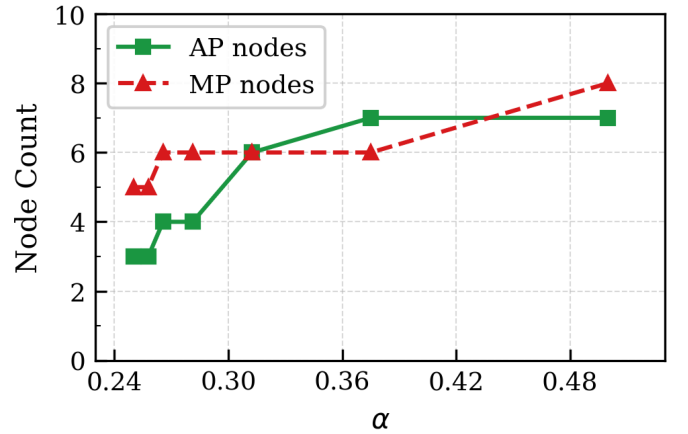


Fig. 3. Active AP and MP node counts as a function of α (QA solver). Both counts are non-decreasing across all iterations.

B. Effect of α on Deployment Density

Fig. 2 plots the total number of active pairs as a function of α for the QA solver. As expected from Section III-E, increasing α strengthens the importance term in (8), so the solver selects more pairs and the count rises monotonically. Fig. 3 breaks this down into AP and MP node counts; both are non-decreasing across all seven binary search iterations, with no budget violation observed in either solver. The feasibility boundary lies between $\alpha = 0.265625$ and $\alpha = 0.375$; the binary search correctly identifies this and converges to $\alpha^* = 0.2578125$ for QA and $\alpha^* = 0.25$ for SA.

C. Localization Accuracy

Table II and Fig. 5 report mean localization error and standard deviation over all 100 seeds. SA achieves 2.69 ± 0.08 m with 5.8 ± 1.3 pairs on average, while QA achieves 2.66 ± 0.09 m with 5.8 ± 1.4 pairs. The gap between QA and SA is modest (≈ 0.03 m) but consistent across seeds, indicating that both solvers reliably find near-optimal placements under this formulation.

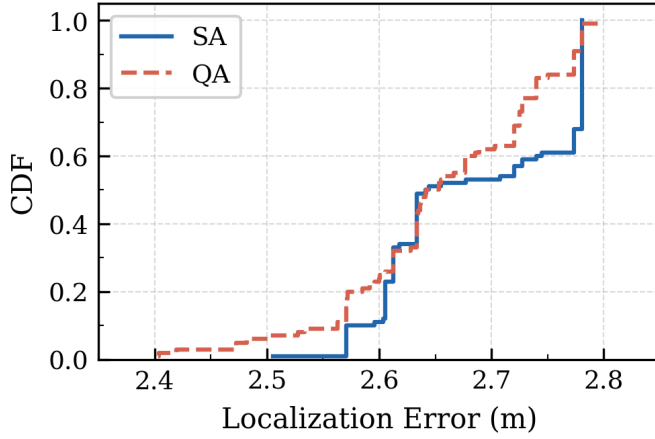


Fig. 4. Empirical CDF of localization error over 100 independent runs for SA and QA. QA achieves a lower best-case error (2.40 m versus 2.51 m for SA) and lower mean, though both distributions overlap in the upper tail.

TABLE II
MULTI-RUN STATISTICS OVER 20 INDEPENDENT SEEDS

Method	Accuracy (m)	Min / Max (m)	Pairs	TTS ₉₉ (ms)
SA	2.69 ± 0.08	2.51 / 2.78	5.8 ± 1.3	2.58 ± 1.42
QA	2.66 ± 0.09	2.40 / 2.80	5.8 ± 1.4	0.020 (fixed)

Fig. 4 shows the empirical CDF of localization error across all 100 runs for SA and QA. QA achieves a lower best-case error of 2.40 m versus 2.51 m for SA, reflecting QA’s ability to discover better deployments in favorable seeds. SA’s worst case is slightly lower (2.78 m versus 2.80 m for QA), and both distributions show comparable spread, with the two CDFs crossing at the upper tail. QA’s advantage is concentrated in the lower half of the distribution, where it more frequently finds high-quality deployments.

Fig. 6 shows hardware utilisation at α^* for a representative run: SA activates 4 APs and 5 MPs, while QA activates 3 APs and 5 MPs. Across all 100 runs, SA uses on average 3.6 APs and 4.7 MPs; QA uses 3.7 APs and 4.6 MPs. Both solvers regularly leave part of the node budget unused, which reflects the redundancy penalty steering the objective away from budget-saturating deployments.

D. Time-to-Solution

Both solvers achieved $p_s = 1.0$ across all reads in every run, so $\text{TTS}_{99} = \tau$ throughout. Fig. 7 shows the per-solve TTS for a representative run: SA requires 1.286 ms (measured wall-clock), while the projected D-Wave anneal time is $\tau = 20 \mu\text{s}$, a $64\times$ speedup for that particular run. Across 100 runs, SA mean TTS_{99} is 2.58 ± 1.42 ms (variation is due to host CPU load); QA TTS_{99} is always exactly 0.020 ms because $p_s = 1.0$ and τ is a fixed hardware constant. The mean per-solve speedup is therefore $129\times$.

Fig. 8 reports total optimization time for a representative run, defined as the cumulative anneal time across all 7 binary search iterations plus the final TTS_{99} at α^* . SA totals 934 ms;

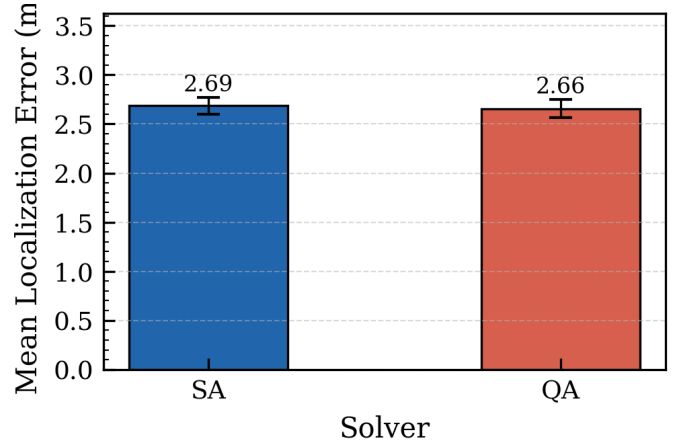


Fig. 5. Mean localization error ($\pm$ one standard deviation) over 20 independent runs for SA and QA. QA achieves a lower mean with a comparable spread.

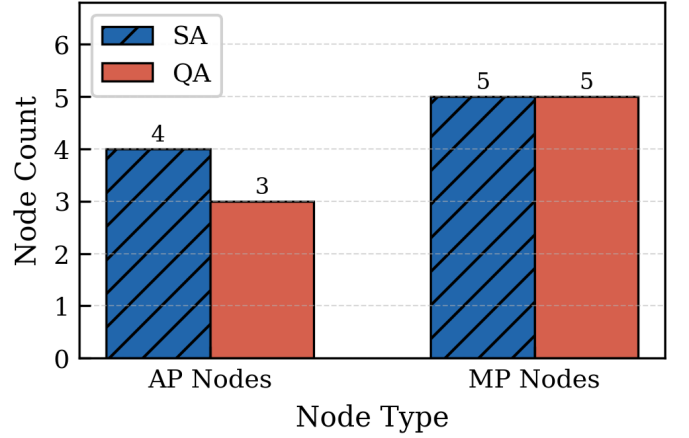


Fig. 6. Active AP and MP node counts at α^* for SA and QA (representative run). Both solvers leave part of the budget unused, guided by the redundancy term in the objective.

the projected QA total is $7 \times 20 \mu\text{s} + 20 \mu\text{s} = 0.16$ ms, a $5,839\times$ reduction. These projections assume that D-Wave programming and readout overhead is amortised and that anneal time dominates, consistent with reported D-Wave specifications [23].

V. RELATED WORK

A. Classical Node placement

Early work on RTI node placement relied on geometric coverage arguments, placing nodes so that their Fresnel zones tile the monitored area as uniformly as possible [2]. More recent heuristic approaches rank links by their individual importance and add them greedily until a budget is reached, but these methods ignore correlations between links and can saturate budgets with redundant measurements [1]. Optimization-based methods have also been proposed, including mixed-integer programming formulations, but these become computationally intractable as the candidate pool grows and do not exploit the

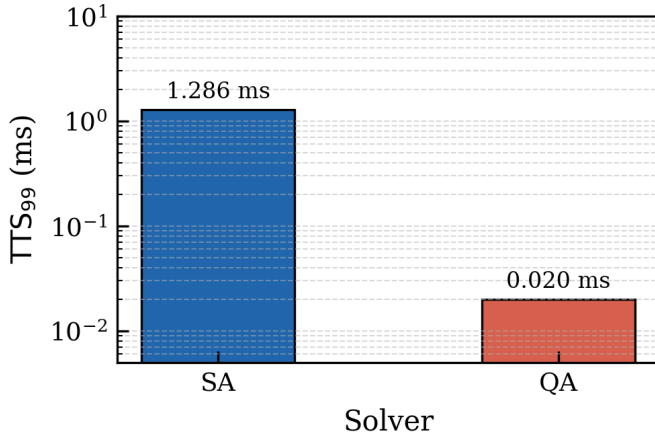


Fig. 7. TTS₉₉ per QUBO solve for SA (measured) and QA (projected D-Wave hardware, $\tau = 20 \mu\text{s}$). Log scale.

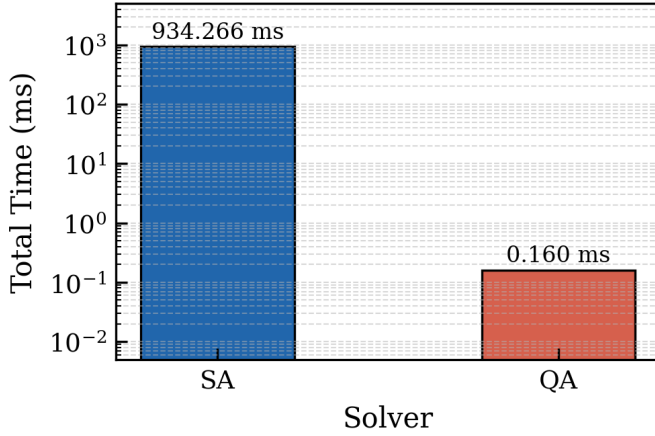


Fig. 8. Total optimization time across all binary search iterations plus TTS₉₉ at α^* , for SA and QA. Log scale.

structure of the problem for efficient solving. Our work differs by casting placement as a QUBO that simultaneously rewards importance and penalizes redundancy, and by solving it with quantum annealing.

xxx

B. Quantum Node Placement

QUBO formulations have been applied to several problems in wireless networking, including access point selection [4], channel assignment, and network topology optimization [27]. The standard way to handle budget constraints in a QUBO is through a quadratic penalty term [3], but this requires careful tuning of the penalty coefficient η for each new instance and can cause the solver to trade localization quality for feasibility. Shokry and Youssef [4] applied QUBO with a penalty term to access point selection for indoor localization, demonstrating the viability of quantum annealing for this class of problems and motivating the penalty-free approach we develop here.

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VI. CONCLUSION

We addressed the problem of AP-MP node placement for device-free localization using quantum annealing. The placement problem is both combinatorially hard and quality-sensitive: the right subset of links must be chosen not just to satisfy hardware budgets on the number of AP and MP nodes, but to maximize spatial diversity and minimize redundancy across the deployed set. Our QUBO formulation captures both objectives directly from RSS fingerprint data, and a provably correct binary search over a single scalar α enforces the AP and MP budgets without any penalty coefficient tuning.

Evaluated on the RTI Indoor Basement dataset over 100 independent runs, the quantum annealing solver achieves a lower mean localization error than simulated annealing and a substantially better best-case result, while using a comparable number of active pairs. The key driver of this improvement is the redundancy term in the objective: without it, solvers tend to saturate both node budgets with correlated links; with it, the search converges to a smaller, spatially diverse set that covers the monitored area more effectively. Projected D-Wave hardware reduces per-solve TTS₉₉ by a mean factor of $129\times$ and total search time by $5,839\times$ relative to simulated annealing, indicating that quantum annealing can make the placement search practical even as the candidate pool grows.

Although we evaluated on an RTI dataset, the formulation generalizes to any link-based DFL system that produces per-link RSS fingerprints. Several directions remain open. The current evaluation relies on projected rather than measured quantum hardware timings; running the placement search on physical D-Wave hardware would validate the speedup claims under real noise conditions. An alternative QUBO representation that operates over node-level binary variables rather than directed pairs would yield a smaller problem matrix and better scaling to larger candidate pools. Extending the framework to online deployments, where nodes can be reconfigured in response to observed localization error, is a natural next step for adaptive RTI systems.

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